



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PRE-SCORE
20-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

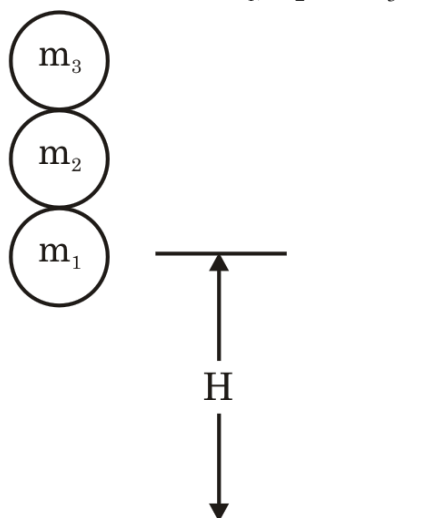
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Three spherical balls having mass m_1 , m_2 and m_3 are placed one on top of another with their center of masses aligned as shown in the figure below. These balls are dropped together from height H . The radii of balls are negligibly small in comparison with height H . The heights to which the masses will rebound is expressed as $H_r = XH$. Assuming that all collisions (including the collision between m_1 and the floor) are elastic and $m_1 \gg m_2 \gg m_3$, the values of X for masses m_1 , m_2 and m_3 are given by



(A) (1, 9, 49)

(B) (1, 9, 27)

(C) (1, 1, 1)

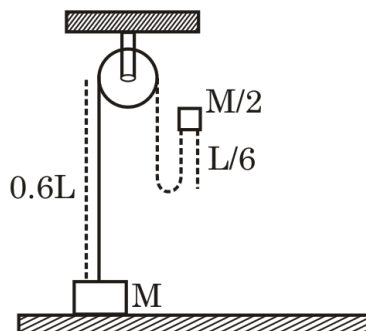
(D) (1, 4, 9)

Space for Rough Work

2. Three objects, a hollow cylinder, a solid cylinder and a solid sphere are released from the top of an inclined plane. The objects, having equal masses and radii, roll without slipping while coming down the plane. Arrange these objects in the descending order of time required to reach the bottom of the plane (slowest to fastest).
- (A) All three objects will take the same time to reach the bottom.
- (B) Hollow cylinder, solid cylinder, solid sphere.
- (C) Hollow cylinder, solid sphere, solid cylinder.
- (D) Solid cylinder, hollow cylinder, solid sphere.
3. A disk shaped coin is tossed up in the air with 'Head' side on the top. The initial centre of mass velocity of the coin is 6.8 m.s^{-1} in the vertical direction. In addition, the disk is rotating about the horizontal axis with angular velocity $6\pi \text{ rad.s}^{-1}$. Assume the motion in air is torque-free. Find the number of completed flips the disk will perform before it hits the ground under the action of gravity.
- (A) 5 completed flips with 'Tail' side on the top.
- (B) 4 completed flips with 'Tail' side on the top.
- (C) 5 completed flips with 'Head' side on the top.
- (D) 4 completed flips with 'Head' side on the top.

Space for Rough Work

4. A block of mass M at rest on the floor is attached to a ball of mass $\frac{M}{2}$ via a light inextensible string of length L which passes over a light small frictionless pulley. The pulley is at a height of $0.6L$ from the floor. The ball is raised to a height of $\frac{L}{6}$ and released (see figure). The speed of the block at the instant the string gets taut (not slack) will be :-



(A) $\frac{\sqrt{2gL/3}}{3}$

(B) $\frac{\sqrt{gL/6}}{3}$

(C) $\frac{\sqrt{4gL}}{3}$

(D) $\frac{\sqrt{1.2gL}}{3}$

Space for Rough Work

5. A particle performs simple harmonic motion. A large number of snapshots of its position are taken at random. A histogram of these positions will :-
- (A) have a minimum around the mean position
- (B) have a maximum around the mean position.
- (C) resemble a sinusoidal curve.
- (D) be almost flat.
6. An explosion releasing energy E occurs in a medium of density ρ . This results in a spherical blast wave of radius r . One can express the radius of the blast wave at time t after the explosion in a form $r = kt^\alpha E^\beta \rho^\gamma$ where k is a dimensionless constant. Then the radial speed of the blast wave can be written as $v \propto t^n$. The value of n is
- (A) -3 (B) $\frac{1}{5}$ (C) $-\frac{3}{5}$ (D) 2

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

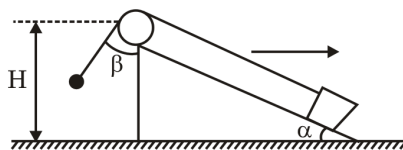
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

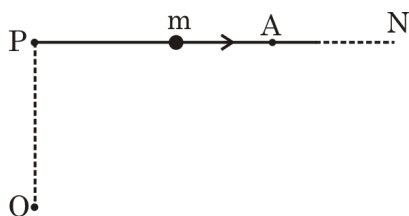
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A wedge with an inclination angle α ($\cos \alpha = \frac{5}{13}$) is placed on a horizontal table. Through a weightless block fixed on a wedge, a light inextensible thread is taken over it, to the ends of which a ball of mass m and a block of mass $\frac{m}{2}$ are tied (see figure). Initially, the system is held motionless by placing the ball at a distance H from the table, while the thread does not sag. Then the wedge began to move with constant horizontal acceleration, and the ball was released. The block and the ball started to move, while the thread tied to the ball made an angle ($\cos \beta = \frac{3}{5}$) with the vertical. All points of the system move in the vertical plane. Neglect the friction in between the axis of the block and the wedge.



- (A) The acceleration of the wedge is $\frac{4g}{3}$.
- (B) The acceleration of block relative to the wedge is $\frac{38g}{39}$.
- (C) Time for the ball to reach the table is $\sqrt{\frac{65H}{19g}}$.
- (D) Time for the ball to reach table is $\sqrt{\frac{57H}{19g}}$.

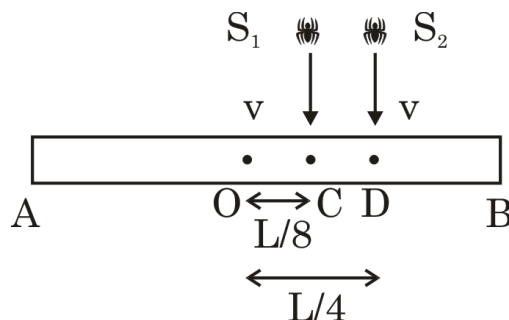
8. The coefficient of static friction between a small block and the surface of a horizontal turntable is 0.09. The block is placed at a distance of 0.1 m from the centre of the turntable. At time $t = 0$ s the turntable starts rotating from rest with a constant angular acceleration of $0.5 \text{ rad}\cdot\text{s}^{-2}$. The time at which the block starts sliding is t_s . Select the correct statement (s) :-
- (A) The angle between the net linear acceleration and radial vector is $\frac{\pi}{4}$ radians.
- (B) t_s is 6 s.
- (C) At t_s the angular speed of the turntable is $6 \text{ rad}\cdot\text{s}^{-1}$.
- (D) The turntable has rotated through 9 radians.
9. A particle of mass m is moving on a straight path (PAN) with constant speed as shown on the right. Let ω and α denote the magnitude of the angular velocity and angular acceleration of the particle with respect to the point O which is not on the path. Then :-



- (A) $\alpha = 0$ and torque on the particle is zero.
- (B) $\alpha \neq 0$ but torque on the particle is zero.
- (C) α increases as the particle moves from P to N.
- (D) ω decreases as the particle moves from P to N.

Space for Rough Work

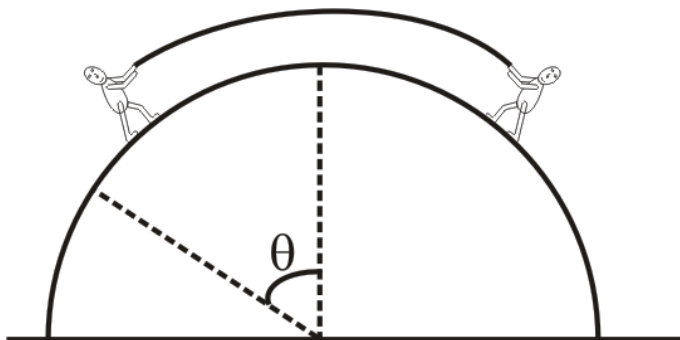
10. A homogeneous rod AB of length L and mass M is pivoted freely at the centre O in such a way that it can rotate freely in the vertical plane as shown in figure. The rod is initially in the horizontal position. Now there are two insects S_1 and S_2 of same mass M falls vertically with equal speed V on points C and D respectively. Points C and D are at a distance of $L/8$ and $L/4$ respectively from O . Immediately after falling the both insects moves towards the end B such that the rod rotates with a constant angular velocity ω and both the insects are moving with the same speed with respect to rod. During the time insects S_2 moved to the end B , the rod was found to rotate by $\frac{\pi}{2}$ rad. Mark the correct statement (s) taking acceleration due to gravity as g .



- (A) Angular speed ω of the rod will be $\frac{72}{31} \frac{V}{L}$.
- (B) The ratio of speeds of one insect at times $t = 0$ to $t = \frac{\pi}{6\omega}$ with respect to the rod is $2 : 1$.
- (C) As insects move towards the end B , during the rotation of rod by $\frac{\pi}{2}$ rad, for the system of rod + insects, we can use the equation " $\tau = I\alpha$ " about a horizontal axis passing through O and perpendicular to the rod.
- (D) The value of V can be found in terms of g , L from the information provided.

Space for Rough Work

11. A tourist of mass 60 kg is roped to a guide of mass 80 kg on the side of a mountain, the side of which may be taken to be an inverted hemisphere. The length of the rope subtends an angle 90° at the centre of the mountain, and the rope is not supposed to touch the mountain at any point. If the coefficient of friction between either man and the mountain is 0.5, in which angular position θ from the centre will the tourist be in equilibrium? Assume that the tourist, guide and rope are in the same vertical plane containing the radius of the hemisphere.



(A) 45°

(B) 60°

(C) 37°

(D) 15°

Space for Rough Work

12. A particle with an initial kinetic energy E_0 approaches the origin from $x = +\infty$ in the presence of a conservative potential. Consider following expressions for the corresponding potential energy $V(x)$ where x is in meters :

$$(i) V(x) = E_0 \left[1 + \exp \left\{ - \left(\frac{x}{x_0} \right)^2 \right\} \right]$$

$$(ii) V(x) = E_0 \left[1 + 1.5 \exp \left\{ - \left(\frac{(x-5)}{x_0} \right)^2 \right\} \right]$$

$$(iii) V(x) = E_0 \left[1 - 0.5 \exp \left\{ - \left(\frac{(x-5)}{x_0} \right)^2 \right\} \right]$$

$$(iv) V(x) = \frac{E_0}{\left(\frac{(x-10)}{x_0} \right)^2} \left[1 - 0.5 \exp \left\{ - \left(\frac{(x-5)}{x_0} \right)^2 \right\} \right]$$

Here $x_0 = 1\text{m}$. Select the correct statement(s)

- (A) The particle will reverse its direction on reaching $x = 5\text{ m}$ for the potential energy given by expression (ii).
- (B) The particle will reach $x = 10\text{ m}$ for the potential energy given by expression (iv).
- (C) The particle will reach the origin for the potential energy given by expression (iii).
- (D) The speed of the particle will decrease monotonically as the particle approaches the origin for the potential energy given by expression (i).

Space for Rough Work

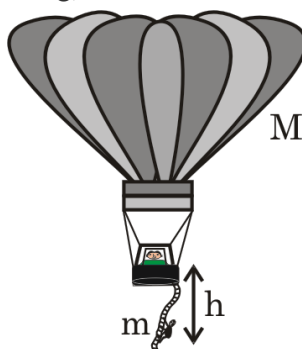
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

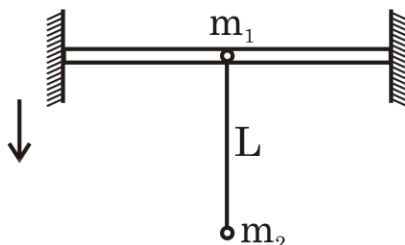
Zero Marks : 0 In all other cases.

1. Mass of balloon + ladder system is M and that of the man is ' m '. Initially the system is floating in air at rest. Find the work done (in J) by internal forces of the man in raising himself up the ladder. (Assume the process is slow) Take $M = 80 \text{ kg}$ & $m = 50 \text{ kg}$, $h = 1 \text{ m}$



Space for Rough Work

2. A ball is projected vertically upwards with a velocity 20 m/s returns back on the ground with a velocity 16 m/s. If the air resistance is proportional to the speed of the ball, find the airtime of the ball. Acceleration of free fall is 10 SI units.
3. Two point masses m_1 and m_2 are connected to each other by a massless and inextensible string of length L mass m_1 is restricted to move along a smooth horizontal rod as shown in the figure. All surfaces are smooth. Find angular frequency (in rad/s) of small oscillations. ($m_1 = 4.4$ kg, $m_2 = 10$ kg, $\ell = 2.5$ m)



4. A car in one dimensional motion with initial speed v_0 is accelerated with constant acceleration a_0 for a distance s . Each time the car travels a distance $s = \frac{v_0^2}{2a_0}$, the acceleration of the car is reduced by a multiplying factor $\lambda < 1$. If the final speed of the car doubles (at a very large distance from the starting point) then the value of λ is.

Space for Rough Work

5. A certain quaternary star system consists of a central star of mass M and three smaller stars, each of mass m . The three smaller stars orbit the central star in the same circular orbit of radius r and have the same sense of revolution. They are positioned one-third of a revolution apart from one another in the orbit. Find the period of revolution of each smaller star (in year).
(r = radius of earth orbit around sun, M = mass of sun, $m = \frac{M}{\sqrt{3}}$)
6. A satellite GeoSAT is in a circular geostationary orbit of radius R_G , above a point P on the equator. Another satellite ComSAT is in a lower circular orbit of radius $0.81 R_G$. At 7 P.M on January 1. ComSAT is sighted directly above P . If both are revolving in the same sense, after how much time (in hours) will COM sat be seen above P ?

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Which of the following ore consists of more than one metal.
 (A) Ilmenite (B) Kieserite (C) Feldspar (D) Both (A) and (C)
2. The metal is available in native form and used in ornamental purpose mainly. For its electrorefining, the electrolyte used is :-
 (A) MCl_3 (aqueous) (B) Molten MCl_3
 (C) MNO_3 (aqueous) (D) Molten M_2O
3. Iodine-azide test can be given by :
 Set-I : Cl^- , Br^- , I^-
 Set-II : S^{2-} , $\text{S}_2\text{O}_3^{2-}$, SCN^-
 Set-III : $\text{S}_2\text{O}_3^{2-}$, SCN^- , CN^-
 Set-IV : S^{2-} , SCN^-
 (A) Set-I only (B) Set-II only
 (C) Set-II and Set-III only (D) Set-II and Set-IV

Space for Rough Work

4. $M^{2+} + Na_2C_2O_4$ solution + AcOH $\rightarrow$ No ppt.

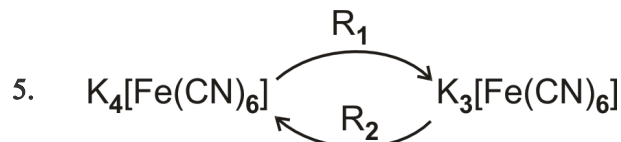
M^{2+} is/are :-

(A) Ca^{2+} , Ba^{2+}

(B) Ca^{2+} , Sr^{2+}

(C) Sr^{2+} , Ba^{2+}

(D) Ca^{2+} , Sr^{2+} , Ba^{2+}



R_1 and R_2 are respectively.

(A) O_3 and H_2O_2/H^+

(B) H_2O_2 (neutral medium) and H_2O_2/OH^-

(C) H_2O_2/OH^- , H_2O_2/H^+

(D) H_2O_2/H^+ , H_2O_2/OH^-

6. The number of identical faces in B_{12} unit is :-

(A) 5

(B) 10

(C) 15

(D) 20

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

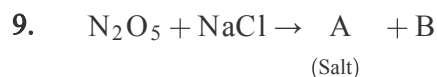
7. Which of the following statement is **INCORRECT**?

- (A) In the smelting step of Cu-extraction, reduction takes place
- (B) True reducing agent for self reduction reaction is PbS for Pb-extraction
- (C) In the electrorefining of Zn, cathode used is strip of pure Zn
- (D) In the roasting step of Cu, all sulphides are converted into their oxides

8. The common metal(s) present in gun metal and bronze is/are :-

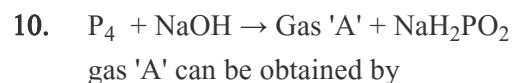
- (A) Cu
- (B) Zn
- (C) Sn
- (D) Fe

Space for Rough Work



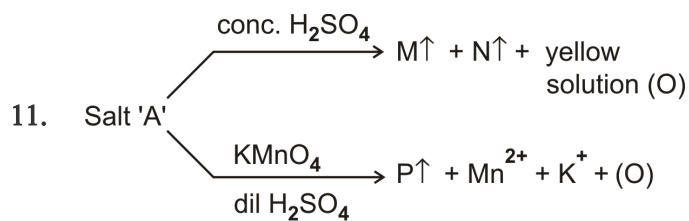
Which of the following characteristics are correct for A and B?

- (A) Hybridisation state of central atom of B and of anionic part of A is same
- (B) Oxidation state of central atom of anionic part of 'A' and that of B is same
- (C) Anionic part of A and B are isoelectronic (considering total electrons)
- (D) Anionic part of A and B are isosteric (considering valence shell electrons only)



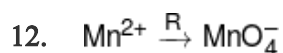
- | | |
|--|--|
| (A) Heating of H_3PO_4 | (B) Heating of H_3PO_3 |
| (C) Heating of H_3PO_2 | (D) Heating of NaH_2PO_2 |

Space for Rough Work



The correct statement regarding above information given :

- (A) Difference in oxidation state of element other than O-atom is 2 in M and N
- (B) MnO_4^- oxidises cationic part of 'A'
- (C) MnO_4^- oxidises anionic part of 'A'
- (D) Gas P and gas N may be same



R is/are :-

- (A) $\text{K}_2\text{S}_2\text{O}_8/\text{H}^+$
- (B) $\text{NaBiO}_3/\text{H}^+$
- (C) $\text{Pb}_2\text{O}_3/\text{HNO}_3$ conc.
- (D) $\text{PbO}_2/\text{conc. HCl}$

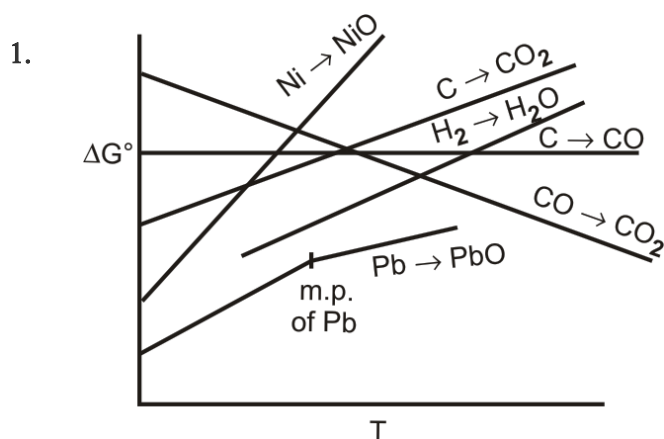
Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.



If number of correct curve is x
and number of incorrect curve is y
then find out the value of x/y .

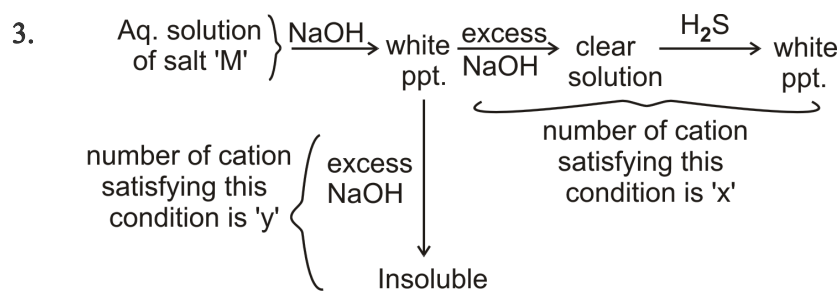
Space for Rough Work

2. For azurite, from the given statements
- (i) It produces bluish green flame in the flame test
 - (ii) On heating it produces red residue
 - (iii) The gas comes out on heating turns lime water milky
 - (iv) After dissolving in dil. HCl, KCl saturated solution is added and final solution is green in colour
 - (v) It is blue in colour

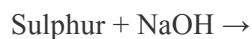
If number of correct statement is x

and number of incorrect statement is y

then find out the value of $\left(\frac{x}{y} + \frac{y}{x}\right)$



Find the value of (y/x)



The sum of oxidation states of Mn, S and F in the products is _____.

Space for Rough Work

5. Find the number of incorrect match.

Compound/molecules		Indicated property
(i)	NH ₃	(MnSO ₄ + H ₂ O ₂) : used for its detection
(ii)	H ₂ S	Conc. H ₂ SO ₄ : Drying agent
(iii)	NH ₃	P ₂ O ₅ : Drying agent
(iv)	O ₃	Turpentine oil : Absorber
(v)	CO	Cu ₂ Cl ₂ (aq.) : Absorber
(vi)	CO	PdCl ₂ solution : used for detection
(vii)	I ₂	Starch : used for detection
(viii)	PH ₃	Rotten fish smell : used for detection

6. The sum of oxidation state of elements other than O-atom in the following ores.
Dolomite, Cryolite, Pyrolusite, Zinc blende, Chromite ore

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $(1 + 2x + 3x^2)^{15} = a_0 + a_1x + a_2x^2 + \dots + a_{30}x^{30}$, then $\sum_{k=1}^{30} k \cdot a_k$ is-
 (A) $20 \cdot 6^{15}$ (B) 6^{15} (C) $20 \cdot 3^{15}$ (D) 3^{15}
2. Mr. X walk up 16 steps, going up either 1 or 2 steps at a time. There is an explosive material on the 8th step so he cannot step there. The number of ways in which Mr. X can go up 16 steps is
 (A) 441 (B) 420 (C) 462 (D) 400
3. If r & s be the positive integers and $f(r, s)$ be the function which gives number of ordered quadruples (a, b, c, d) of positive integers such that $3^r 7^s = \text{LCM}(a, b, c) = \text{LCM}(a, b, d) = \text{LCM}(a, c, d) = \text{LCM}(b, c, d)$, then
 (A) $f(1,1) = 122$ (B) $f(2,1) = 364$ (C) $f(1,2) = 364$ (D) $f(3,3) = 4489$

Space for Rough Work

4. Let $xy = 2$

$$\Delta = \begin{vmatrix} 1+x^2-y^2 & 2xy & -2y \\ 2xy & 1-x^2+y^2 & 2x \\ 2y & -2x & 1-x^2-y^2 \end{vmatrix},$$

then minimum value of Δ is

- (A) 28 (B) 8 (C) 3 (D) 125

5. Let $P = [p_{ij}]$ be a 3×3 non singular matrix and a matrix $Q = [q_{ij}]$ is formed such that $q_{ij} = \sum_{\substack{r=1 \\ r \neq j}}^3 p_{ir}$ then

value of $\det(P^{-1}Q)$ is

- (A) 2 (B) 0 (C) 1 (D) -1

6. Let $a, b, c, x, y, z \in \mathbb{R}$ such that

$$ax + by + cz = 1$$

$$bx + cy + az = 2$$

$$cx + ay + bz = 3.$$

then the value of $(a^3 + b^3 + c^3 - 3abc)(x^3 + y^3 + z^3 - 3xyz) =$

- (A) 18 (B) -18 (C) 9 (D) -9

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Consider the word 'INSTITUTE'

(A) Number of ways of selection of 4 letters is 41

(B) Total number of words beginning with vowel is $2 \times \frac{8!}{3!}$

(C) Total number of words which are formed by 4 letters are 626

(D) Total number of words beginning with vowel is $\frac{8!}{3}$

Space for Rough Work

8. Consider all possible strings of n digits which can be formed using digits from set $\{1, 2, 3, 4\}$. Let E_n denotes number of strings having even number of 1's whereas O_n denotes number of strings having odd number of 1's. Identify the correct option(s)

(A) $E_n = \frac{4^n + 2^n}{2}$

(B) $O_n = \frac{4^n + 2^n}{2}$

(C) $E_n = \frac{4^n - 2^n}{2}$

(D) $O_n = \frac{4^n - 2^n}{2}$

9. Consider a sequence of 10A's and 8 B's placed in a row. By a run we mean single or more letters of the same type placed side by side. For example following is an arrangement consisting of 4 runs of A & 4 runs of B : AAABBABBBBAABAAAABB. The number of arrangements of 10 A's and 8 B's in a row so that there are -

(A) 4 runs of A and 4 runs of B is $2 \cdot {}^9C_3 \cdot {}^7C_3$

(B) 4 runs of A and 5 runs of B is $3 \cdot {}^9C_4 \cdot {}^7C_4$

(C) 3 runs of A and 3 runs of B is $2 \cdot {}^9C_2 \cdot {}^7C_2$

(D) 6 runs of A and 6 runs of B is $4 \cdot {}^9C_5 \cdot {}^7C_5$

Space for Rough Work

10. Let α, β are real numbers for which the system of linear equations
 $2\alpha x + y + z = 0$
 $\alpha x + y + 2z = 1$
 $\beta^2 x + y - 3\alpha z = -2$
 never posses unique solution then point $P(\alpha, \beta)$ lies on a conic whose-
- (A) center is $\left(-\frac{1}{2}, 0\right)$ (B) eccentricity is less than 3
 (C) one vertex is origin (D) $(-1, 0)$ lies on it
11. Let $f(p, q) = (2p + q)^7$, $h(r) = \left(r^2 - \frac{2}{r}\right)^9$ and $S(p, q, r) = f(p, q).h(r)$, then which of the following is/are correct ?
- (A) Coefficient of $p^4 q^3$ in expansion of $S(p, q, r)$ is $2^{12}.3^1.5^1.7^2$
 (B) Greatest coefficient in expansion of $S(p, q, r)$ is $2^{13}.3^2.7^2$
 (C) Number of terms in expansion of $S(p, q, r)$ is 80
 (D) Sum of coefficient in expansion of $S(p, q, r)$ is -2187
12. Consider the number $N = (10)!$, then
- (A) Number of divisors of N is 270
 (B) Number of divisors of N , divisible by $5!$, is 96
 (C) Sum of all odd divisors of N is 30008
 (D) Number of divisors which are perfect cube is 16.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Value of $1 \cdot {}^{20}C_0 \cdot {}^{30}C_0 + 2 \cdot {}^{20}C_1 \cdot {}^{30}C_1 + 3 \cdot {}^{20}C_2 \cdot {}^{30}C_2 + \dots + 21 \cdot {}^{20}C_{20} \cdot {}^{30}C_{20}$ is ${}^{50}C_{20} + 20 \cdot {}^{49}C_q$ (where q is even number), then $\frac{q}{q+1}$ is equal to
 2. We have 19 identical gems available with us which are needed to be distributed among A, B and C such that A always gets an even number of gems. If the number of ways this can be done is K then find the value of $\left(\frac{K}{88}\right)$?
 3. If number of arrangements of letters of the word "DHARAMSHALA" taken all at a time so that no two alike letters appear together is $(4^a \cdot 5^b \cdot 6^c \cdot 7^d)$, (where a, b, c, d $\in$ N), then a + b + c + d is equal to
-

Space for Rough Work

4. If $c < 2$ and the system of equation $x + y - 2z = 0$, $2x + y - cz = 0$ and $-bx + 2by - cz = 0$ has non-trivial solution then if range of b is (m, n) then value of $\left| \frac{1}{n - m} \right|$ is
5. Number of value(s) of λ for which $\exists a, b, c \in \mathbb{R}$ satisfying
 $(1 - \lambda)a + 3b - 4c = 0$
 $a - (3 + \lambda)b + 5c = 0$
 $3a + b - \lambda c = 0$
and $a^2 + b^2 + c^2 \neq 0$, is 'x' then $\frac{x}{3}$ is
6. Let B is adjoint of matrix A , having order 3 and $B^T B^{-1} = A$ (where B is non singular), then $\frac{\text{tr}(A + B)}{4}$ is
(where $\text{tr}(A)$ is sum of diagonal elements of matrix A)
-

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator**Space for Rough Work**